

Harsh Gupta

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Summary

AI Engineer with an MS in Data Science and 3+ years delivering production GenAI – multi-tenant RAG platforms, NLP pipelines, and LLM-powered decision systems on Azure and AWS. Skilled in LangChain, LangGraph, Python, and end-to-end delivery from retrieval architecture and model fine-tuning to stakeholder-facing AI solutions.

Education

Indiana University Bloomington

Master of Science in Data Science (GPA: 4.0/4.0)

Manipal University

B.Tech in Electronics and Communication, Minor in Data Science (GPA: 8.13/10)

Aug 2024 – May 2026

Bloomington, IN

Jul 2017 – May 2021

Manipal, India

Technical Skills

Programming: Python, PySpark, SQL, FastAPI, JavaScript

Generative AI & NLP: LangChain, LangGraph, MCP, RAG, Vector Search, Prompt Engineering, Text Classification, Sentiment Analysis, Azure AI Studio, Hugging Face, Ollama

Machine Learning & MLOps: PyTorch, Time Series Forecasting, Model Fine-tuning, NLP, Model Deployment, Docker, CI/CD

Cloud & Data Platforms: AWS (Glue, S3, Redshift), Azure, Postgres, OracleDB, PowerBI

Experience

Indiana University Bloomington

Part-Time AI/ML Engineer | Azure, Python, LangChain, LLM, RAG, NLP

Oct 2024 – Present

Bloomington, IN

- Built and deployed **Chat AIU**, a multi-tenant RAG platform powering chatbots on **10+ IU department websites**, with self-service knowledge ingestion from Confluence, documents, sitemaps, and URLs.
- Improved retrieval quality via **semantic reranking** and LLM evaluation pipelines; extended the platform with **MCP, A2A, and OpenAPI agent integrations** for tool-augmented conversational AI.
- Built analytics for chat sessions, prompts, and flagged content – giving department stakeholders visibility into chatbot performance and knowledge gaps.
- Building an **ML/LLM decision support system** for parking appeals – classifying violations, extracting policy clauses, and generating reviewer rationale via fine-tuned **SmolLM (135M)** – cutting adjudication time from weeks to days.

Indiana University Bloomington Campus Auxiliaries

AI/ML Intern | Azure ML, Python, Time Series, Scikit-learn

Jun 2025 – Aug 2025

Bloomington, IN

- Analyzed **millions of dining transaction records** to build time-series forecasting models for footfall prediction, reducing average forecast error from **~2,000 swipes to ± 500 swipes per day**.
- Partnered with operations and finance stakeholders to translate model outputs into actionable food preparation and labor scheduling strategies.
- Drove an estimated **\$2K–\$3K daily operational cost optimization** through reduced waste and overstaffing via model-driven scheduling.

Deloitte – Strategy & Analytics

Data Engineer & Data Scientist | Python, AWS Glue, PySpark, LangChain, R, ML

Sep 2021 – Jul 2024

Bengaluru, India

- Built nightly batch ETL pipelines in **AWS Glue and PySpark** processing **12–15M insurance records** into a **Landing → Staging → ODS** architecture with row-level audit tracking and reusable transformation modules ensuring data lineage at scale.

Data Scientist – Customer Strategy & Pricing | Python, R, ML, AWS S3, Tableau

- Developed a **linear mixed-effects regression model** across **5,000+ locations** to optimize pricing strategy, increasing profit by **\$1M–\$3M** while limiting churn to **1%**; presented findings directly to executive stakeholders.

Firm Initiative | LangChain, PySpark, Flask, Postgres, AWS S3

- Shipped **DataLens** (LLM-powered RAG for enterprise NL querying) and **CodETL** (ETL engine, **40% dev effort reduction**); presented at the **Deloitte AI & DE Summit**, earning **4+ firm awards**.

Projects

Multi-Agent, Voice-Enabled Interview Prep Platform | LLM, LangChain, LangGraph, Python, Hugging Face

- Built a **multi-agent AI interview simulator** using LangGraph to orchestrate specialized agents – question generation, response evaluation, and adaptive feedback – across role-specific technical interview scenarios.
- Added a **voice-enabled interview mode** using speech-to-text and text-to-speech models for fully spoken mock interviews with real-time response handling.

TA-Lite | Python, Ollama, SQL, LangChain

- Developed a modular **AI teaching assistant** using local LLMs (Ollama) and RAG over course materials to deliver context-aware, hint-based academic support with instructor-configurable prompt personas.
- Designed with a layered retrieval architecture separating chunking, retrieval, and prompt assembly – enabling instructors to adjust hint depth and topic scope without code changes.